NIHAL HUSSAIN

Coimbatore, Tamil Nadu

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SUMMARY

Computer Science and Engineering student specializing in AI/ML, Agentic AI, and intelligent application development. Experienced in building AI-powered applications using Python, machine learning, NLP, LLMs, and Retrieval-Augmented Generation (RAG), alongside full-stack development. Strong problem-solving background with 2,850+ algorithmic problems solved across competitive programming platforms.

EDUCATION

B.E – Computer Science and Engineering (Pre-Final Year) — KIT-KALAIGNARKARUNANIDHI INSTITUTE OF TECHNOLOGY

CGPA: 8.42/10 — **2024-2028**

PROJECTS

LaunchCrew (React.js Monaco Editor, Python, FastAPI, MongoDB, Multi-Agent Orchestration, Tavily, Playwright, SSE, JWT, pytest) (Demo)

- Engineered an 12-agent AI platform that transforms natural-language requirements into websites through coordinated research, planning, design, code generation, testing, and deployment workflows.
- Developed an AI-powered IDE with Monaco Editor, code diffs, version control, deterministic code intelligence, change-impact analysis, and automated debugging.
- Implemented Playwright-based QA, security and accessibility audits, human-in-the-loop approvals, JWT authentication, live agent traces, and Docker deployment; validated with 1,145 pytest tests.

CivicAINexus(React.js, MongoDB, Python, Machine Learning, NLP, RAG, Speech-to-Text, AI Agents) (Demo)

- Developed a multilingual AI grievance platform enabling citizens to submit government complaints through text, images, and voice in four languages, with role-based dashboards for citizens and officials.
- Developed NLP/ML pipelines for complaint classification, department routing, priority prediction, sentiment analysis, duplicate detection, and summarization, reaching 99.92% accuracy on 6300 test complaints.
- Built an agentic AI workflow that autonomously assists with grievance creation, complaint analysis, information retrieval, and response generation using RAG-based knowledge retrieval,while integrating role-based dashboards for citizens and government officials.

AI Phishing Detector (Python, Flask, Scikit-learn, HTML, CSS, JavaScript, Machine Learning, MongoDB) (Demo)

- Developed an AI-powered phishing website detection system that analyzes URLs using machine learning to identify malicious websites in real time.
- Engineered a feature extraction pipeline and trained a Scikit-learn classification model, integrating it into a Flask application for accurate phishing prediction.
- Built a responsive web interface using HTML, CSS, and JavaScript and integrated MongoDB to securely store user queries, prediction history, and application data.

EXPERIENCE

DECODELABS, Remote Internship (Certificate)

- Applied Python, Pandas, NumPy, Scikit-learn, and Machine Learning concepts to design and develop AI-based solutions, including rule-based chatbots, recommendation systems, and data classification models, while following structured development practices.
- Strengthened practical expertise in data preprocessing, feature engineering, model development, algorithm implementation, Git/GitHub version control, and debugging, contributing to the successful completion of multiple AI/ML projects in a collaborative learning environment.

TECHNICAL SKILLS AND CERTIFICATIONS

TECHNICAL SKILLS

Programming Languages: C++, Python, C, JavaScript, SQL

AI/ML & Agentic AI: Machine Learning, Scikit-learn, NumPy, Pandas, LLMs, Prompt Engineering, RAG, NLP, AI Workflows

Web Development & Frameworks: Flask, React.js, Node.js, HTML, CSS, REST APIs

Databases, Tools & Core Concepts: MySQL, MongoDB, Git, GitHub, Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems (OS)

CERTIFICATIONS

- Principles of Generative AI Certification — Link
- Python Foundation Certification — Link
- Artificial Intelligence Fundamentals — Link
- Prompt Engineering Basics — Link
- Claude Code 101 — Link
- fundamentals of generative ai and large language models — Link
- Digital marketing and e-commerce — Link

COMPETITIVE PROGRAMMING

LeetCode: Current Rating: 1924| Solved Problems: 900+ | Top 4.9% Globally | Highest Rating 1924 | Knight Badge |

Codeforces: Current Rating: 1060 | Solved Problems: 300+ | Highest Rating : 1071 |

CodeChef: Current Rating: 1570 | Position: 2 Star | Solved Problems: 900+ | Highest Rating: 1608 | Global Ranking: 15676 |

ACHIEVEMENTS

- Participated in a 36-hour innovation and software development challenge, collaborating to design and develop an AI-powered food waste tracking solution.
- Solved 2850+ algorithmic and data structure problems across competitive programming platforms, participated in 310+ coding contests, and maintained a 680+ day problem-solving streak.
- Achieved a 5-star rating on HackerRank, reflecting proficiency in problem-solving and core programming concepts.